LEARNER GUIDE

Equation of a Straight Line

Straight-Line Graphs. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

A straight line has an equation of the form $y = mx + c$. The two numbers in it tell a story: m (the gradient) is how fast y changes, and c (the y-intercept) is where the story starts. This lesson helps you read m and c from a line, write the equation, and say what a line means in a real situation. The question you are exploring is: can a line tell the story of change?

What you are learning to notice

- That in $y = mx + c$, m is the gradient (rate of change) and c is the starting value (the y-intercept).
- How to read m and c from a graph, a table or a description.
- What positive, negative and zero gradients look like.
- How to write the equation of a line and explain what it says about a real situation.

Moving through it, one step at a time

Use the explorer to change m and c and watch the line move — steeper, gentler, up or down, starting higher or lower. Match equations to tables, graphs and stories. When you meet a real context, ask what the gradient and the starting value mean in that story. Take your time and try predictions before you check them.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (negative numbers, the idea that every point on the y-axis has $x = 0$, and substitution), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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